Julia Nonnenkamp

Computer Science Ph.D. Student at UW–Madison interested in systems for human-centered security and mitigations for technology-facilitated abuse.

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Education

2024 – Present **University of Wisconsin–Madison** – Madison, WI
Ph.D. in Computer Science *(Focus: security & privacy)*
Advisor: Dr. Rahul Chatterjee

2020 – 2024 **Georgetown University** – Washington, DC
B.A. in Computer Science & Math
Research mentors: Dr. Elissa Redmiles, Dr. Lisa Singh

Research experience

July 2024 – Present **Ph.D. Student, UW–Madison**
Advisor: Dr. Rahul Chatterjee
Research Assistant (Spring 2024 onward), Independent Study (Fall 2024)

- Demonstrated the feasibility of using GDPR/CCPA data exports to identify past or ongoing account compromise, with a focus on intimate partner violence *(CCS '25)*; developed tooling to standardize and extract security information from data exports so that users can navigate and annotate their own data *(under submission)*.
- Supported a qualitative user study to understand how adults with intellectual and developmental disabilities (IDD) interact with everyday security and privacy notifications *(ASSETS '26)*; building a tool informed by the study's findings *(ongoing)*.
- Supported building and evaluating an agentic assistant that automates security checks for devices and accounts, focusing on survivors of technology-facilitated abuse *(under submission)*.

August 2023 – May 2024 **Undergraduate Researcher, Georgetown University**
Massive Data Institute & Georgetown Law
Research mentors: Dr. Elissa Redmiles, Dr. Lisa Singh
Associated with Fritz Family Fellowship

- Examined law enforcement materials about software systems, algorithms, and verification procedures utilized in forensic DNA analysis. Built synthetic data generation pipelines to audit security risks for forensic DNA analysis software systems.

Publications

Under Submission **LEStrADE: Investigating Account Compromise using Data Exports for Survivors of Technology-Facilitated Abuse.** Julia Nonnenkamp, Naman Gupta, Angela Wang, Sophie Stephenson, Rahul Chatterjee.

Under Submission — **Clinicas: Agentic Diagnosis and Mitigation of Technology-Facilitated Abuse.**
Shirley Zhang, Guruprasad Viswanathan Ramesh, Julia Nonnenkamp, Rahul Chatterjee, Kassem Fawaz.

October 2026 — **"I'm trying not to get hacked:" How Adults with Intellectual and Developmental Disabilities Navigate Security and Privacy Notifications**
Hailey L. Johnson, Julia Nonnenkamp, Bilge Mutlu, Rahul Chatterjee.
ACM Computers and Accessibility (ASSETS). October 2026.

October 2025 — **Hidden in Plain Bytes: Investigating Interpersonal Account Compromise with Data Exports.**
Julia Nonnenkamp, Naman Gupta, Abhimanyu Dev Gupta, Rahul Chatterjee.
ACM Computer and Communications Security (CCS). October 2025.

Talks & posters

August 2026 — Poster: Hidden in Plain Bytes: Investigating Interpersonal Account Compromise with Data Exports.
GREPSEC doctoral consortium at the USENIX Security Symposium

February 2026 — Flash Talk: The Madison Tech Clinic
Tech for Good Conference at the University of Chicago

October 2025 — Talk: Investigating Interpersonal Account Compromise with Data Exports
ACM Computer and Communications Security (CCS) Conference

Throughout 2025-6 — Various "Security & the News" talks (e.g., age verification, forced decryption)
Weekly MadS&P Seminar

April 2024 — Talk: The Technology behind Forensic Analysis
Fritz Family Fellows Conference

April 2024 — Poster: Auditing Probabilistic Genotyping Software used in Forensic DNA Analysis
Massive Data Institute Research Showcase

December 2023 — Poster: Risk Mapping in Forensic DNA Analysis
Massive Data Institute Research Showcase

Honors and awards

2026 — GREPSEC Student Travel Grant, USENIX Security Symposium

2026 — Finalist, Aspen Institute Science and Technology Policy Fellowship

2025 — Student Conference Travel Grant (ACM CCS)

2023 – 2024 — Fritz Family Fellowship (Georgetown University)
Interdisciplinary research fellowship through GU's Tech & Society Initiative.

Related work experience

Summer 2024 **Two Six Laboratories — Research Intern** – Arlington, VA
Applied hybrid program analysis techniques to identify anomalous behavior in mobile messaging applications using Frida, Python, Haskell, and SMT solvers.

June 2023 – May 2024 **U.S. Office of Management & Budget – Intern** – Washington, DC
Evaluated experimental applicability of transformer-based architectures for time-series forecasting of budget data. Refactored, tested, and documented econometric models in Python to improve code readability. Produced accessible, interactive visualizations for historical Presidential Budget data in JS and D3.

2022 – 2024 **Students of Georgetown, Inc. (The Corp)** – Washington, DC
- *Director of Data Analytics (2023-24)*: Built databases and ETL pipelines for point-of-sale data (PostgreSQL, Mage.ai, Python). Produced data communications and visualizations for store leads.
- *Director of Operations, People Operations (2022-23)*: Ran payroll for 300+ employees, led Federal Work-Study partnership, managed company-wide onboarding.

Teaching experience

Fall 2024 **Teaching assistant, CS 220: Data Science Programming** – UW–Madison

Fall 2022 **Student academic assistant, COSC 160: Data Structures (C++)** – Georgetown

2021 – 2022 **Private Calculus (I-III) Tutor** – Washington, DC

2020 – 2021 **Instructor, Mathnasium** – Arlington, VA

Service and outreach

2025 – Present **Mad S&P Seminar – Co-leader**

2024 – Present **Madison Tech Clinic – Technology Consultant, Volunteer Coordinator**
MTC is a volunteer partnership between UW—Madison and local intimate partner violence (IPV) advocacy groups and shelters. Through one-on-one consultations, we provide trauma-informed cybersecurity assistance to IPV survivors, helping them identify points of compromise and strengthen their security practices.

Technical skills

Reverse Engineering & Verification
Frida, Haskell, SMT solvers (Z3), Ghidra (learning, very novice), Rocq/Coq (novice).

Data Engineering & Analysis
Python (pandas/numpy, PyTorch), SQL, Mage.ai, Google Timesketch, OpenSearch (novice).

General
C++, Java, Flask, JavaScript (D3.js, Vue.js).